

JLSP01-4G

Sleep Monitoring Device - Communication Integration Document

Product Manual (Ver. 1.3) Reference

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MQTT Communication Data Format Description (V1.3)

1. Description

This document mainly describes the MQTT protocol format of the smart sleep/respiration radar device. It helps users understand how to obtain and set radar data for convenient development use. Data content is transmitted in JSON format.

The smart sleep/respiration radar device has two topics, as follows:

(1) Topic: /Radar60SP/device IMEI/sys/property/post

Reporting topic. The radar device reports data and setting response messages through this topic.

(2) Topic: /Radar60SP/device IMEI/sys/property/set

Downlink topic. The customer publishes messages to the radar device through this topic.

2. Radar Function Properties

Online/Offline (online)

Actively reported. The property status is divided into: 0 (offline status), 1 (online status). It is actively reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "online": "0" // Device offline
  }
}
```

Heartbeat Packet (heartBeat)

Actively reported. The property status is divided into: 0 (abnormal), 1 (normal). It is actively reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "heartBeat": "0"
  }
}
```

Device Software Version (firmwareVersionProduct)

Actively reported. The data length is 28 bytes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "firmwareVersionProduct": "JLSP01-CAT1_2024.6.5_V1.0.6"
  }
}
```

This property supports query. The query format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "firmwareVersionProduct": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success", // fail on failure; success on success
  "params": {
    "firmwareVersionProduct": "JLSP01-CAT1_2024.6.5_V1.0.6"
  }
}
```

```
}
```

Device Radar Sensor Software Version (firmwareVersion)

Actively reported. The data length is 15 bytes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "firmwareVersion": "xxxx"
  }
}
```

This property supports query. The query format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "firmwareVersion": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success", // fail on failure; success on success
  "params": {
    "firmwareVersion": "XXXX"
  }
}
```

Cat.1 Communication Module Software Version (firmwareVersionWiFi)

The device actively reports this when it is powered on and connected to the server. The specific format is as follows:

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "firmwareVersionCat1": "xxxxxxx"
  }
}
```

This property supports query. The query format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "firmwareVersionCat1": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success", // fail on failure; success on success
  "params": {
    "firmwareVersionCat1": "XXXX"
  }
}
```

Location Out-of-Bounds Status (locationOutOfBounds)

Actively reported. The property value range is divided into: 0 (outside the range), 1 (inside the range). It is reported when the status changes.

```
{
  "version": "1.0",
```

```
"method": "post",
"params": {
  "locationOutOfBounds": "0"
}
}
```

This property supports query. The query format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "locationOutOfBounds": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success", // fail on failure; success on success
  "params": {
    "locationOutOfBounds": "1"
  }
}
```

Human Detection Function Switch (humanSwitch)

Does not support active reporting. The range is divided into: 1 (function enabled), 0 (function disabled).

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "humanSwitch": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "humanSwitch": "1"
  }
}
```

This property supports setting. The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "humanSwitch": "1"
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success" // fail on failure; success on success
  "params": {
    "humanSwitch": "1"
  }
}
```

Human Presence Status (someoneExists)

Actively reported. The property value range is 0 (no person), 1 (person present). It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "someoneExists": "1" // Someone is currently present in the environment
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "someoneExists": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "someoneExists": "1"
  }
}
```

Human Motion Information (motionStatus)

Actively reported. The property value range is 0 (none), 1 (stationary), 2 (active). It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "motionStatus": "1" // Stationary status
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "motionStatus": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "motionStatus": "1"
  }
}
```

Movement Sign Parameter (movementSigns)

Actively reported. The property value range is 0-100. Data is reported periodically.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "movementSigns": "100"
  }
}
```

```
}
```

Human Distance (humanDistance)

Actively reported. The property value range is 0-65535, unit: cm. Data is reported periodically.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "humanDistance": "100"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "humanDistance": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "humanDistance": "1"
  }
}
```

Human Position (humanPosition)

Actively reported. The property value range is x-axis: 0-65535, y-axis: 0-65535, z-axis: 0-65535. Data is reported periodically.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "humanPosition": {
      "x": "xxx",
      "y": "xxx",
      "z": "xxx"
    }
  }
}
```

This property supports query. Currently, only simultaneous query of x, y, and z is supported, in the following format:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "humanPosition": {"x": "?", "y": "?", "z": "?"}
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "humanPosition": {
      "x": "xxx",
      "y": "xxx",
      "z": "xxx"
    }
  }
}
```

Breathing Function Switch (breathSwitch)

Does not support active reporting. The range is divided into: 1 (function enabled), 0 (function disabled).

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "breathSwitch": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "breathSwitch": "1"
  }
}
```

This property supports setting. The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "breathSwitch": "1" // Set to enabled
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success" // fail on failure; success on success
  "params": {
    "breathSwitch": "1"
  }
}
```

Breathing Information (breathInform)

Actively reported. The property value range is 1 (normal), 2 (breathing too high), 3 (breathing too low), 4 (detecting). It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "breathInform": "1"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "breathInform": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
```

```
    "breathInform": "1"
  }
}
```

Breathing Value (breathValue)

Actively reported. The property value range is 0-35. It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "breathValue": "1"
  }
}
```

Breathing Intensity (breathIntensity) - Not Available Yet

Actively reported. The property value range is 0-100. It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "breathIntensity": "1"
  }
}
```

Breathing Waveform (breathWave)

Actively reported. Each report contains one group of data, consisting of 5 integer values, with a range of 0-255. Reported periodically.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "breathWave": [128, 128, 128, 128, 128]
  }
}
```

Heart Rate Function Switch (heartRateSwitch)

Does not support active reporting. The range is divided into: 1 (function enabled), 0 (function disabled).

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "heartRateSwitch": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "heartRateSwitch": "1"
  }
}
```

This property supports setting. The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "heartRateSwitch": "1" // Set to enabled
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success" // fail on failure; success on success
  "params": {
    "heartRateSwitch": "1"
  }
}
```

Heart Rate Value (heartRateValue)

Actively reported. The property value range is 60-120. It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "heartRateValue": "1"
  }
}
```

Heart Rate Waveform (heartRateWave)

Actively reported. Each report contains one group of data, consisting of 5 integer values, with a range of 0-255. Reported periodically.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "heartRateWave": [128, 128, 128, 128, 128]
  }
}
```

Sleep Function Switch (sleepSwitch)

Does not support active reporting. The range is divided into: 1 (function enabled), 0 (function disabled).

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "sleepSwitch": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "sleepSwitch": "1"
  }
}
```

This property supports setting. The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "sleepSwitch": "1" // Set to enabled
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success" // fail on failure; success on success
}
```

```
"params": {
  "sleepSwitch": "1"
}
```

Sleep Quality Status (sleepStatus)

Actively reported. The property value range is: 0 (deep sleep), 1 (light sleep), 2 (awake), 3: none (reported when out of bed).

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "sleepStatus": "1"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "sleepStatus": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "sleepStatus": "1"
  }
}
```

In-Bed/Out-of-Bed Status (getIntoBed)

Actively reported. The property value range is 0 (out of bed), 1 (in bed). It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "getIntoBed": "1"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "getIntoBed": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "getIntoBed": "1"
  }
}
```

Awake Duration (awakeDuration)

Actively reported. The property value range is 0-65536. It is reported when the status changes. Unit: minutes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "awakeDuration": "2345"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "awakeDuration": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "awakeDuration": "2345"
  }
}
```

Light Sleep Duration (lightSleepDuration)

Actively reported. The property value range is 0-65536. It is reported when the status changes. Unit: minutes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "lightSleepDuration": "1"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "lightSleepDuration": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "lightSleepDuration": "1"
  }
}
```

Deep Sleep Duration (deepSleepDuration)

Actively reported. The property value range is 0-65536. It is reported when the status changes. Unit: minutes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "deepSleepDuration": "1"
  }
}
```

This property supports query. The format is as follows:

```
{
```

```
"version": "1.0",
"method": "get",
"params": {
  "deepSleepDuration": "?"
}
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "deepSleepDuration": "1"
  }
}
```

Single Sleep Segment End Score (sleepScore)

Actively reported. The property value range is 0-100. It is reported when the status changes.

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "sleepScore": "80"
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "sleepScore": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success" // fail on failure; success on success
  "params": {
    "sleepScore": "80"
  }
}
```

Sleep Detection / Real-Time Detection Mode Selection (reportMode)

The range is divided into: 1 (sleep status transmission), 0 (real-time data transmission). The default is sleep status.

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "reportMode": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success",
  "params": {
    "reportMode": "1"
  }
}
```

This property supports setting. The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "reportMode": "1"
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success" // fail on failure; success on success
  "params": {
    "reportMode": "1"
  }
}
```

Single Sleep Segment Comprehensive Status (sleepComprehensiveStatus)

Actively reported. Each report contains one group of data, consisting of 8 integer values, with a range of 0-255. Reported periodically. The data meanings in order are:

[1B presence, 1B sleep status, 1B average respiration, 1B average heartbeat, 1B number of turns, 1B proportion of large body movements, 1B proportion of small body movements, 1B number of apnea events]

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "sleepComprehensiveStatus": [1, 1, 18, 19, 18, 32, 20, 2]
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "sleepComprehensiveStatus": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success",
  "params": {
    "sleepComprehensiveStatus": [1, 1, 18, 19, 18, 32, 20, 2]
  }
}
```

Single Sleep Segment End Quality Analysis (sleepQualityAnalysis)

Actively reported. Each report contains one group of data, consisting of 12 integer values, with a range of 0-255.

The data meanings in order are: [1B sleep quality score, 2B total sleep duration, 1B awake duration ratio, 1B light sleep duration ratio, 1B deep sleep duration ratio, 1B out-of-bed duration, 1B number of times out of bed, 1B number of turns, 1B average respiration, 1B average heartbeat, 1B number of apnea events]

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "sleepQualityAnalysis": [1, 2, 52, 35, 24, 32, 32, 69, 16, 36, 86, 32]
  }
}
```

This property supports query. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "sleepQualityAnalysis": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success",
  "params": {
    "sleepQualityAnalysis": [1, 2, 52, 35, 24, 32, 32, 69, 16, 36, 86, 32]
  }
}
```

Abnormal Sleep Status (abnormalSleep)

Actively reported. The range is divided into: 0 (sleep duration less than 4 hours), 1 (sleep duration greater than 12 hours), 2 (abnormally no person for a long time), 3 (none).

```
{
  "version": "1.0",
  "method": "post",
  "params": {
    "abnormalSleep": "1"
  }
}
```

Voice Function Switch (US518FunctionSwitch)

After this function is turned off, the device will not perform voice inquiries or voice interaction. The sensor detection status is still reported normally.

This property supports setting and query, does not support active reporting, where 1 means enabled and 0 means disabled. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "US518FunctionSwitch": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success"
  "params": {
    "US518FunctionSwitch": "1"
  }
}
```

Set the username for connecting the device to the server (not more than 16 bytes). The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "US518FunctionSwitch": "0"
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success"
  "params": {
    "US518FunctionSwitch": "0"
  }
}
```

```
}  
}
```

Voice Learning Mode (US518LearningFunction)

Note: To set this property, someone must be present; otherwise it cannot be set correctly, and the reply value is "nobody".

Does not support active reporting. The range is divided into: 1 (enter learning mode), 0 (not in learning mode). The default status is 0.

Note: After voice learning is triggered, the learning process must be executed and completed before it can be triggered again. Temporarily, it can only be triggered once within 60 seconds. When the 60-second period ends, the status is actively reported.

This property supports query. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "get",  
  "params": {  
    "US518LearningFunction": "?"  
  }  
}
```

The query result format is as follows:

```
{  
  "version": "1.0",  
  "opt": "get",  
  "res": "success"  
  "params": {  
    "US518LearningFunction": "1"  
  }  
}
```

This property supports setting. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "set",  
  "params": {  
    "US518LearningFunction": "1"  
  }  
}
```

The setting result format is as follows:

```
{  
  "version": "1.0",  
  "opt": "set",  
  "res": "success"  
  "params": {  
    "US518LearningFunction": "1" // If someone is present, reply 1 or 0; if no one is present, reply  
nobody  
  }  
}
```

Night Light Control Switch (nightLightSwitch)

Actively reported. The property value range is: 1 (turn on the light), 0 (turn off the light). The format is as follows:

```
{  
  "version": "1.0",  
  "method": "post",  
  "params": {  
    "nightLightSwitch": "1"  
  }  
}
```

This property supports query. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "get",  
  "params": {  
    "nightLightSwitch": "?"  
  }  
}
```

```
}  
}
```

The query result format is as follows:

```
{  
  "version": "1.0",  
  "opt": "get",  
  "res": "success"  
  "params": {  
    "nightLightSwitch": "1"  
  }  
}
```

This property supports setting. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "set",  
  "params": {  
    "nightLightSwitch": "1"  
  }  
}
```

The setting result format is as follows:

```
{  
  "version": "1.0",  
  "opt": "set",  
  "res": "success"  
  "params": {  
    "nightLightSwitch": "1"  
  }  
}
```

Get Server Information (getServer)

Note: The factory default address of the device is "tcp://broker.emqx.io", with port 1883, unless there are additional requirements.

Does not support active reporting.

This property supports query. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "get",  
  "params": {  
    "getServer": "?"  
  }  
}
```

The query result format is as follows:

```
{  
  "version": "1.0",  
  "opt": "get",  
  "res": "success"  
  "params": {  
    "getServer": "broker.emqx.io", "1883"  
  }  
}
```

Configure Server Information (setServer)

Note: To set the server, write "tcp://broker.emqx.io",1883. Escape characters \" are required before and after the path address.

Does not support active reporting.

This property supports setting. The format is as follows:

```
{  
  "version": "1.0",  
  "method": "set",  
  "params": {  
    "setServer": "broker.emqx.io", "1883"  
  }  
}
```

```
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success"
  "params": {
    "setServer": "broker.emqx.io", "1883"
  }
}
```

Query Device IMEI Number (deviceIMEI)

This property supports query and does not support active reporting. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "deviceIMEI": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success"
  "params": {
    "deviceIMEI": "xxxxxxxx"
  }
}
```

Query Device SIM Card Number (deviceSIM)

This property supports query and does not support active reporting. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "deviceSIM": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success"
  "params": {
    "deviceSIM": "xxxxxxxx"
  }
}
```

Username for Connecting to Server (username)

This property supports setting and query and does not support active reporting. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "username": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success"
}
```

```
"params": {
  "username": "xxxxxxx"
}
```

Set the username for connecting the device to the server (not more than 16 bytes). The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "username": "Enter the username to be set"
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success"
  "params": {
    "username": "The username that was set"
  }
}
```

Password for Connecting to Server (password)

This property supports setting and query and does not support active reporting. The format is as follows:

```
{
  "version": "1.0",
  "method": "get",
  "params": {
    "password": "?"
  }
}
```

The query result format is as follows:

```
{
  "version": "1.0",
  "opt": "get",
  "res": "success"
  "params": {
    "password": "xxxxxxx"
  }
}
```

Set the password for connecting the device to the server (not more than 16 bytes). The format is as follows:

```
{
  "version": "1.0",
  "method": "set",
  "params": {
    "password": "Enter the password to be set"
  }
}
```

The setting result format is as follows:

```
{
  "version": "1.0",
  "opt": "set",
  "res": "success"
  "params": {
    "password": "The password that was set"
  }
}
```

3. Restrict Function Property Reporting

This function mainly targets function properties in actively reported data that are reported frequently. It prohibits their reporting, or masks certain function properties so that they are not reported.

For the corresponding property value, 1 means reporting of this property is prohibited, and 0 means reporting is allowed.

Setting commands with method "limit_set" or "limit_get" can set multiple properties.

The format for querying "prohibited reporting data" is as follows:

```
{
  "version": "1.0",
  "method": "limit_get",
  "params": {
    "locationOutOfBounds": "?",
    "someoneExists": "?",
    "motionStatus": "?",
    "movementSigns": "?",
    "humanDistance": "?",
    "humanPosition": "?",
    "heartBeat": "?",
    "breathValue": "?",
    "heartRateValue": "?"
  }
}
```

The response result data format is as follows:

```
{
  "version": "1.0",
  "opt": "limit_get",
  "res": "success" // fail on failure; success on success
  "params": {
    "locationOutOfBounds": "0",
    "someoneExists": "0",
    "motionStatus": "0",
    "movementSigns": "0",
    "humanDistance": "0",
    "humanPosition": "0",
    "heartBeat": "0",
    "breathValue": "0",
    "heartRateValue": "0"
  }
}
```

The format for setting "prohibited reporting data" is as follows:

```
{
  "version": "1.0",
  "method": "limit_set",
  "params": {
    "locationOutOfBounds": "0",
    "someoneExists": "0",
    "motionStatus": "0",
    "movementSigns": "1",
    "humanDistance": "1",
    "humanPosition": "1",
    "heartBeat": "0",
    "breathValue": "0",
    "heartRateValue": "0"
  }
}
```

The response result data format is as follows:

```
{
  "version": "1.0",
  "opt": "limit_set",
  "res": "success" // fail on failure; success on success
  "params": {
    "workDuration": "0",
    "locationOutOfBounds": "0",
    "someoneExists": "0",
    "motionStatus": "0",
    "movementSigns": "1",
    "humanDistance": "1",
    "humanPosition": "1",
    "heartBeat": "0",
    "breathValue": "0",
    "heartRateValue": "0"
  }
}
```

4. Set QoS Service Quality

The default QoS service quality is QoS1. Value 0 indicates QoS0, 1 indicates QoS1, and 2 indicates QoS2.

The function for querying QoS service quality is as follows:

```
{
  "version": "1.0",
  "method": "qos_get",
  "params": {
    "Qos_post": "?"
  }
}
```

The response query result data format is as follows:

```
{
  "version": "1.0",
  "opt": "qos_set",
  "res": "success" // fail on failure; success on success
  "params": {
    "Qos_post": "0"
  }
}
```

The function for setting QoS service quality is as follows:

```
{
  "version": "1.0",
  "method": "qos_set",
  "params": {
    "Qos_post": "0"
  }
}
```

The response result data format is as follows:

```
{
  "version": "1.0",
  "opt": "qos_set",
  "res": "success" // fail on failure; success on success
  "params": {
    "Qos_post": "0"
  }
}
```

5. OTA Upgrade

Upgrade Module (otaModule)

This property requires a command to be issued to distinguish the module that needs to be upgraded. 0 indicates the Wi-Fi module, and 1 indicates the radar module. The default is 0. This control command must be issued before each OTA upgrade.

```
{
  "version": "1.0",
  "method": "ota_set",
  "params": {
    "module": "0"
  }
}
```

Upgrade Information (otaInform)

The issued upgrade information includes the firmware address and port number. After the device receives this data, it obtains the upgrade firmware from the specified URL and port number, and then starts the upgrade. If it fails, it returns failure. After success, the module restarts and reports the upgraded version number. The customer must build the web server by themselves.

```
{
  "version": "1.0",
  "method": "ota_set",
  "params": {
    "otaInform": {
      "firmwareUrl": "http://192.168.31.88/app.rbl",
      "port": "8090" // Port number
    }
  }
}
```

```
}
}
}
```

The return format for upgrade failure is as follows:

```
{
  "version": "1.0",
  "opt": "otaInform",
  "res": "fail",
  "msg": "ota failed!"
}
```

Upgrade Progress (downloadProgress)

During upgrade, the upgrade progress is actively reported. The data range is 0-100. The specific format is as follows:

```
{
  "version": "1.0",
  "method": "ota_post",
  "params": {
    "downloadProgress": "0"
  }
}
```

Historical Version Update Notes

Revision	Release Date	Summary	Author
V1.0	2024/07/26	Initial draft	
V1.1	2024/11/01	Added commands related to server query and setting. Added commands related to voice switch and learning.	
V1.2	2024/11/16	Corrected symbol errors in the port number for obtaining server information and configuring server information.	
V1.3	2025/1/17	Added night light switch command.	